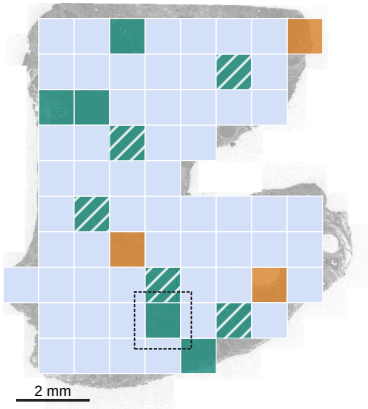
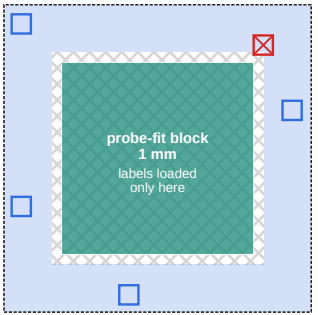
a Split within a section: 1 mm blocks, fixed once



zoom (dashed box in left map)



hatched: 50 µm guard band, molecules dropped; 100 µm training windows (blue) may not reach it (red: rejected)

train 80%
validation 5%
probe-fit 7.5%
probe-test 7.5%

b Held-out ladder: every claim names its rung

harder generalization

4	**a whole platform** platform shift	phase 4
3	**dataset with a new panel** panel shift	phase 4
2	**all sections of a donor** donor variation	phase 3, 5
1	**whole sections, seen donor** section variation	phase 3
0	**1 mm blocks** fit, not generalization	phases 1–2

c Phases: each ends at a gate; a failed gate stops the next phase

0 Data and vocabulary	**1** Pilot on Xenium breast	**2** Panel robustness	**3** Multi-section	**4** Cross-platform transfer	**5** Scale up
vocabulary, CellWorld checkpoints, leakage check; split, baselines	1a frozen CellWorld 1b init from CellWorld 1c from scratch falsification first	on the 1b/1c winner: panel dropout thinning, count noise panel/platform tokens	10–20 sections 2–3 platforms random context size	zero-shot, then platform token, then last block or LoRA	more donors model size; data fractions by section
every gene → one token; leakage status known; split frozen	no collapse; 1b or 1c beats 1a and composition at 2 of 3 scales	≥80% retention on 35%-overlap split; <5% IID cost	beats composition on rungs 1–2	beats composition on shared genes, cell and niche	final numbers on held-out tissue (rungs 2–4)

recipe throughout: AdamW, BF16, batch 256, warmup to 3e-3 then cosine, EMA 0.999 → 0.9999, 10% hint, 10k steps in the pilot (peak LR 1e-3 when starting from CellWorld); block masking at 5/10/50 µm, random masking as first ablation; adaptation only via platform token, last block or LoRA.